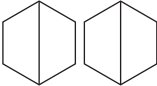


CEPPER

FOR IV/IM



COMPOSITION

Cefer contains sulbactam sodium/cefoperazone sodium combination in a 1:1 in terms of free sulbactam and cefoperazone. The 1:1 vial contains sulbactam sodium equivalent to sulbactam 500 mg and cefoperazone sodium equivalent to cefoperazone 500 mg.

PRODUCT DESCRIPTION

Dry homogeneous white to off-white crystalline powder free from foreign matter filled in colorless vial. Reconstituted solution : Clear, slight yellow to pale amber solution.

PHARMACODYNAMICS

Cefoperazone is 3rd generation cephalosporins, which acts against sensitive organisms during the stage of active multiplication by inhibiting biosynthesis of cell wall mucopeptide. Sulbactam does not posses any useful antibacterial activity except *Neisseriaceae* and *Acinetobacter*. It is an irreversible inhibitor of most important β -lactamases produced by β -lactam antibiotic resistant organisms. The potential for sulbactam's preventing the destruction of penicillins and cephalosporins by resistant organisms. Combination of cefoperazone and sulbactam demonstrates synergistic activity in a variety of organisms, most markedly the following.

Haemophilus influenzae
Bacteroides species
Staphylococcus species
Acinetobacter calcoaceticus
Enterobacter aerogenes
E. coli
Proteus mirabilis
Klebsiella pneumoniae
Morganella morganii
Citrobacter freundii
Enterobacter cloacae
Citrobacter diversus.

Sulbactam/cefoperazone is active against a wide variety orgaisms :

Gram positive : *Staphylococcus aureus*, *penicillinase* and *non-penicillinase producing strain*, *S. epidermidis*, *S. pneumoniae*, *S. Pyrogenes*, *S. agalactiae*, *most strain of Beta-hemolytic streptococci*
Gram negative : *E. coli*, *Klebsiella species*, *Enterobacter species*, *Citrobacter species*, *Haemophilus influenzae*, *Proteus mirabilis*, *Proteus vulgaris*, *Marganella morganii*.

PHARMACOKINETICS

Approximately 84% of sulbactam and 25% of cefoperazone is excreted by the kidney. Most of the remaining cefoperazone is excreted via bile. Half-life of sulbactam is 1 hr while that for cefoperazone is 1.7 hrs. Both cefoperazone and sulbactam distribute well into a variety of tissues and fluids including bile, gall bladder, skin appendix, fallopian tubes, ovary, uterus and others.

INDICATIONS

Cefer is indicated for the treatment of following infections ;

- Upper and lower respiratory tract infections
- Upper and lower urinary tract infections
- Peritonitis
- Cholecystitis
- Cholangitis
- Septicemia
- Meningitis
- Skin and soft tissue infections
- Bone and joint infections
- Pelvic inflammatory disease, endometritis, gonorrhea and other infections of the genital tract.

DOSAGE AND ADMINISTRATION

Adult :

Usual dose : 2-4 g/day of the 1:1 ratio (ie, cefoperazone 1-2 g/day) given IV or IM equally divided dose every 12 hrs.

In severe infection, the daily dose may be increased to 8 g/day of the 1:1 ratio (ie, cefoperazone 4 g/day).

Use in children :

Usual dose is 40-80 mg/kg/day of the 1:1 ratio (ie, cefoperazone 20-40 mg/kg/day) given IV or IM equally divided dose every 6-12 hrs.

In serious or refractory infection, the daily dose may be increased to 160 mg/kg/day of the 1:1 ratio (ie, cefoperazone 80 mg/kg/day).

The recommended maximum daily dose of sulbactam is 4 g.

IV administration

Intermittent injection

Reconstituted each vial with sterile water for injection to volume 4 ml and then dilute to 20 ml followed by administration for 15-60 minutes.

Reconstituted solution should keep in room temperature not exceeding 30°C not more than 24 hrs or in temperature 2-8°C not more than 5 days.

IV injection

Each vial should be reconstituted as mentioned previously and administered over a minimum of 3 minutes.

IM administration

Lidocaine HCl 2% is a suitable vehicle for IM administration, however, not for initial reconstitution.

Dosage in Renal dysfunction :

Patients with creatinine clearance 15-30 ml/min should receive the maximum dose of 1 g sulbactam every 12 hrs. Patients with creatinine clearance <15 ml/min should receive the maximum dose of 500 mg every 12 hrs. In severe infection may be separately adding dose of cefoperazone.

Dosage in Hepatic dysfunction :

Cefoperazone is mainly excreted in bile. Dose modification may be necessary in case of severe biliary obstruction, severe hepatic disease or concomitant with renal dysfunction. In these cases, the dose of cefoperazone should not exceed 2 g/day.

Pregnancy and lactation :

Use in pregnancy

No adequate and well-controlled studies in pregnant women. This drugs should be used during pregnancy only if clearly needed.

Use in nursing mother

Only small quantities of CEPPER are excreted in human milk. Caution should be exercised when the drug was administered in nursing mother.

Use in neonates

It has not been extensively studied in premature infants or neonates. Therefore, the potential benefit and possible risk must be considered before starting this medication. Cefoperazone does not replace bilirubin from plasma protein binding sites.

CONTRAINDICATIONS

Known hypersensitivity to penicillins, cephalosporins, cefoperazone, sulbactam or any component of the formulation.

WARNING

1. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Cefer careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Cefer must be discontinued immediately and appropriate alternative therapy instituted.

2. Do not use in hypersensitive patients.

3. If skin rash, irritation or smelling should discontinued and consult your doctor.

Hypersensitivity : If an allergic reaction occurs, the drug should be discontinued and the appropriate therapy instituted. Serious anaphylactic reaction require immediate emergency treatment with epinephrine, oxygen, intravenous steroids and airway management including intubation should be administered.

PRECAUTION

General

Vitamin K deficiency has occurred in a few patients treated with Cefoperazone. Those at risk include patients with poor diet, malabsorption states (e.g. cystic fibrosis), patients on prolonged intravenous alimentation regimens and patients receiving anticoagulant. The prothrombin time should be monitored and exogenous vitamin K should be administered. Prolonged using may cause overgrowth of non-susceptible organisms. It is advisable to check periodically for organ system dysfunction during extended therapy ; this includes renal, hepatic, and hematopoietic systems. This is particularly important in neonates, especially when premature, and other infants.

Use in hepatic dysfunction

Cefoperazone is extensively excreted in bile. Dosage adjustment may be necessary in cases of severe biliary obstruction, severe hepatic disease or in cases of renal dysfunction coexistent with either of those conditions. In patients with hepatic dysfunction and concomitant renal impairment, dosage should not exceed 2 g of Cefoperazone activity/day, Cefoperazone serum concentrations should be monitored and dosage adjustment as necessary.

Use in renal dysfunction

Patients with creatinine clearances between 15-30 ml/min the maximum dose of Sulbactam should not exceed 1 g every 12 hours, while patients with creatinine clearances less than 15 ml/min the maximum dose of Sulbactam should not exceed 500 mg every 12 hours.

In severe case, the additional dose of Cefoperazone alone may be given. Patients with hemodialysis, half-life of serum Cefoperazone is reduced slightly during hemodialysis. Thus, dosing should be scheduled after dialysis period.

ADVERSE EFFECTS

Cefer is generally well tolerated. The majority of adverse effects are of mild or moderate severity and are tolerated with continued treatment.

Gastrointestinal : Nausea, vomiting, loose stool, Pseudomembranous colitis

Dermatologic Reactions : Maculopapular rash, urticaria

Central nervous system : Hypotension

Hematological : Decrease in neutrophil, neutropenia, decreased hemoglobin or hematocrit, Transient eosinophilia, thrombocytopenia, hypoprothrombinemia

Miscellaneous : Headache, fever, injection pain, chills

Laboratory abnormalities : Transient elevations of liver function tests, SGOT, SGPT, Alkaline phosphatase and bilirubin

Local reactions : transient pain, phlebitis

DRUG INTERACTIONS

- Alcohol : Should not ingested alcohol during and as late as the 5th day after cefoperazone administration. For patients requiring artificial feeding orally or parentally, solutions containing ethanol should be avoided.

- Drug laboratory test interactions : A false-positive reaction of glucose in the urine may occur with Benedict's or Fehling's solution.

OVERDOSAGE

Limited information is available on the acute toxicity of Cefoperazone sodium and Sulbactam sodium in humans. Overdosage of the drug would be expected to product manifestations that are principally extensions of the adverse reactions reported with the drug. The fact that high CSF concentrations of β -lactam antibiotics may cause neurologic effects, including seizures, should be considered. Because Cefoperazone and Sulbactam are both removed from the circulation by hemodialysis, these procedures may enhance elimination of the drug from the body if overdosage occurs in patients with impaired renal function.

INCOMPATIBILITIES

Aminoglycosides

Solution of CEPPER and aminoglycosides should not be directly mixed, Since there is a physical incompatibilities between them. The combination therapy can be accomplished by sequential intermittent intravenous infusion provided that separate secondary intravenous tubing is used, and the primary intravenous tubing is adequately irrigated with an appropriate diluent between doses. It is also suggested that each dose of CEPPER should be administered as far as the dose of aminoglycosides infusion as possible.

Lactated Ringer's solution

Lactated Ringer's solution should not be used for initial reconstituted solution. Sterile water for injection should be used for initial reconstitution then further diluted with Lactated Ringer's solution to obtained Sulbactam concentration of 5 mg/ml

Lidocaine

2% lidocaine should not be used for initial reconstituted solution. Sterile water for injection should be used for initial reconstitution then further diluted with 2% lidocaine to obtained solutions of 0.5% lidocaine.

STORAGE CONDITION

Store at temperature not exceeding 30°C.

Reconstituted solution : Store at room temperature not exceeding 30°C or at temperature 2-8°C.

SHELF-LIFE : 2 years

Reconstituted solution : Store at room temperature not exceeding 30°C not more than 24 hours or at temperature 2-8°C not more than 5 days.

DOSAGE FORM

Powder for injection

PACKAGING AVAILABLE

Individual pack of glass vial 1 g in the box.

Date of revision : April 2019



Manufactured by
BIOLAB CO., LTD.
625 Bangpoo Industrial Estate,
Samutprakarn, Thailand

EEI 042/A

Code : EEI042/A

Draft Revision : 3

Final Artwork Revision : B

Name : AW_LEAFLET CEPPER NEW WARNING MAL.ai

Remark :

Design by

Review by

Approved by

Signature

Signature

Signature

Date

Date

Date

Specification

Size : 190 x.245 MM.(+/- 3 MM.)

Color : พิมพ์ 1 สี คือ สีดำ

Material : ใช้กระดาษปอนด์ น้ำหนัก 55 แกรม

Other :

Note :

#0=เพิ่มความหัวหรือ WARNING
#1=แก้ไข Lactam
#2=เปลี่ยนเป็น Cefer
#3=ปรับลดพิกัดเป็น 30 องศา

☐ ลูกทำผิดพลาดจนผิด ☐ ทำเสนอราคา ☐ ส่งพิมพ์
☒ RA ขึ้นทะเบียน ☐ อื่นๆ.....

COLOR GUIDE



F-00P-QE-011